

GIRRAWEE HIGH SCHOOL

YEAR 11 – TASK4

2009

MATHEMATICS

Time Allowed: 90 minutes

INSTRUCTIONS TO STUDENTS

- Attempt **ALL** questions.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new sheet of paper.

QUESTION 1 (10 marks)

a) Evaluate the following limits:

$$(i) \lim_{x \rightarrow 2} \frac{x+4}{x} \quad 1$$

$$(ii) \lim_{x \rightarrow -5} \frac{x^2 - 25}{x + 5} \quad 2$$

$$(iii) \lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1} \quad 2$$

$$(iv) \lim_{x \rightarrow \infty} \frac{4x^2 + x}{x^2 + 2x + 1} \quad 2$$

b) Differentiate $f(x) = x^2 + 2x + 1$ from first principles. 3

QUESTION 2 (14 marks)

Differentiate:

(i) $y = 3x^4 - 5x^3 + 4x$ 1

(ii) $y = \frac{2x^3 - 4x + 3}{x^2}$ 3

(iii) $y = \sqrt{3x}$ 2

(iv) $y = 4x^{\frac{3}{2}} - 3x^{\frac{7}{3}}$ 2

(v) $y = \frac{7}{\sqrt{x}} + x^3$ 3

(vi) $y = \frac{3}{x^4} + \sqrt[3]{x}$ 3

QUESTION 3 (17 marks)

a) Use the product rule to differentiate $y = (5x^3 + 2)(2x - x^3)$. 3

b) Differentiate the following:

(i) $y = \frac{3x}{x^2 + 3}$ 3

(ii) $y = (x^4 - 2x^3)^5$ 2

(iii) $y = x^3(1 + x)^3$ 3

(iv) $y = \frac{1}{(4x - 2)^3}$ 2

(v) $y = \frac{x}{\sqrt{x^2 + 1}}$ 4

QUESTION 4 (13 marks)

- a) At what point on the curve $y = \frac{x}{x-1}$ is the tangent parallel to the line $x + y + 5 = 0$? 4
- b) At what points on the curve $y = x^3 - x^2$ does the tangent make an angle of 45° with the positive x -axis? 4
- c) The tangent at $P(3,9)$ on the curve $y = x^2$ cuts the x -axis at T , and PN is perpendicular to the x -axis. Find the length of TN . 5

QUESTION 5 (10 marks)

- a) The line $y = mx + b$ is a tangent to the curve $y = x^3 - 3x + 1$ at the point $(-2, -1)$. Find the values of m and b . 4
- b) Find the equations of the *tangent* and *normal* to the curve $y = 2x^3 + 3x - 1$ at the point where $x = 1$. 6

QUESTION 6 (13 marks)

- a) Find the quadratic equation with roots $1 + \sqrt{3}$ and $1 - \sqrt{3}$. 2
- b) For what value(s) of m does the equation $(2m+1)x^2 + 2mx - 1 = 0$ have equal roots? 3
- c) Given that α and β are the roots of the quadratic equation $x^2 - 5x + 4 = 0$, evaluate
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\alpha^2 + \beta^2$ 2
- (iv) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2
- (v) $(\alpha + 2)(\beta + 2)$ 2

QUESTION 7 (15 marks)

a) For what values of k is the function $3x^2 + 8x - k$ positive definite? **3**

b) For what values of k does the quadratic equation

$$x^2 - 5x + (k - 1) = 0 \text{ have}$$

(i) two real roots? **3**

(ii) one real root? **1**

(iii) no real roots? **1**

c) Find the values of A , B and C if

$$4x^2 - 12x + 9 \equiv A(x - 1)^2 + B(x - 1) + C$$
 4

d) Solve: $3^{2x} - 10 \times 3^x + 9 = 0$. **3**

Question 1 (10 marks)

a) i) $\lim_{x \rightarrow 2} \frac{x+4}{x} = 3$ (1)

ii) $\lim_{x \rightarrow -5} \frac{x^2-25}{x+5} = \lim_{x \rightarrow -5} \frac{(x+5)(x-5)}{(x+5)} = \lim_{x \rightarrow -5} (x-5) = -10$ (2)

iii) $\lim_{x \rightarrow 1} \frac{x^2-3x+2}{x-1} = \lim_{x \rightarrow 1} \frac{(x-1)(x-2)}{x-1} = \lim_{x \rightarrow 1} (x-2) = -1$ (2)

iv) $\lim_{x \rightarrow \infty} \frac{4x^2+x}{x^2+2x+1} = \lim_{x \rightarrow \infty} \frac{4+\frac{1}{x}}{1+\frac{2}{x}+\frac{1}{x^2}} = 4$ (2)

b) $f(x) = x^2+2x+1$
 $f(x+h) = (x+h)^2+2(x+h)+1 = x^2+2xh+h^2+2x+2h+1$
 $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = \lim_{h \rightarrow 0} \frac{x^2+2xh+h^2+2x+2h+1-x^2-2x-1}{h} = \lim_{h \rightarrow 0} \frac{2xh+h^2+2h}{h} = 2x+2$

$= \lim_{h \rightarrow 0} 2x+h+2 = 2x+2$ (3)

Question 2 (14 marks)

i) $y = 3x^4-5x^3+4x$
 $\frac{dy}{dx} = 12x^3-15x^2+4$ (1)

ii) $y = \frac{2x^3-4x+3}{x^2} = 2x - \frac{4}{x} + \frac{3}{x^2}$
 $\frac{dy}{dx} = 2 + 4x^{-2} - 6x^{-3} = 2 + \frac{4}{x^2} - \frac{6}{x^3}$ (2)

iii) $y = \sqrt{3x} = (3x)^{1/2}$
 $\frac{dy}{dx} = \frac{1}{2} (3x)^{-1/2} \cdot 3 = \frac{3}{2\sqrt{3x}}$ (2)

iv) $y = 4x^{3/2} - 3x^{7/3}$
 $\frac{dy}{dx} = 6x^{1/2} - 7x^{4/3} = 6\sqrt{x} - 7\sqrt[3]{x^4}$ (2)

v) $y = \frac{7}{\sqrt{x}} + x^3 = 7x^{-1/2} + x^3$
 $\frac{dy}{dx} = -\frac{7}{2} x^{-3/2} + 3x^2 = -\frac{7}{2\sqrt{x^3}} + 3x^2$ (3)

vi) $y = \frac{3}{x^4} + \sqrt[3]{x} = 3x^{-4} + x^{1/3}$
 $\frac{dy}{dx} = -12x^{-5} + \frac{1}{3} x^{-2/3} = -\frac{12}{x^5} + \frac{1}{3\sqrt[3]{x^2}}$ (3)

Question 3 (17 marks)

a) $y = (5x^3+2)(2x-x^3)$
 $\frac{dy}{dx} = vu' + uv' = (5x^3+2)(2-3x^2) + (2x-x^3)(15x^2) = 30x^3 - 15x^5 + 10x^3 - 15x^5 + 4 - 6x^2 = -30x^5 + 40x^3 - 6x^2 + 4$ (3)

b) i) $y = \frac{3x}{x^2+3}$
 $\frac{dy}{dx} = \frac{vu' - uv'}{v^2} = \frac{3(x^2+3) - 3x(2x)}{(x^2+3)^2} = \frac{3x^2+9-6x^2}{(x^2+3)^2} = \frac{9-3x^2}{(x^2+3)^2} = \frac{3(3-x^2)}{(x^2+3)^2}$ (3)

ii) $y = (x^4-2x^3)^5$
 $\frac{dy}{dx} = 5(x^4-2x^3)^4 (4x^3-6x^2) = 5(x^4-2x^3)^4 \cdot 2x^2(2x-3) = 10x^2(x^4-2x^3)^4(2x-3)$ (2)

iii) $y = x^3(1+x)^3$
 $\frac{dy}{dx} = vu' + uv' = 3x^2(1+x)^3 + 3x^3(1+x)^2 = 3x^2(1+x)^2(1+x+x) = 3x^2(1+x)^2(1+2x)$ (3)

iv) $y = \frac{1}{(4x-2)^3} = (4x-2)^{-3}$
 $\frac{dy}{dx} = -3(4x-2)^{-4} \cdot 4 = -\frac{12}{(4x-2)^4}$ (2)

v) $y = \frac{x}{\sqrt{x^2+1}}$
 $\frac{dy}{dx} = \frac{vu' - uv'}{v^2} = \frac{\sqrt{x^2+1} - x \cdot \frac{x}{\sqrt{x^2+1}}}{x^2+1} = \frac{x^2+1 - x^2}{(x^2+1)\sqrt{x^2+1}} = \frac{1}{(x^2+1)^{3/2}}$ (4)

Question 4 (13 marks)

$$c) y = x^2$$

$$\frac{dy}{dx} = 2x$$

$$\text{at } x = 3,$$

$$m = 2(3) = 6; \text{ pt } (3, 9)$$

$$\text{Eq}^n: y - y_1 = m(x - x_1)$$

$$y - 9 = 6(x - 3)$$

$$y - 9 = 6x - 18$$

$$y = 6x - 9$$

$$x \text{ int } \Rightarrow y = 0$$

$$6x - 9 = 0$$

$$x = \frac{9}{6} = \frac{3}{2}$$

$$\therefore \text{Coordinates of T } (1\frac{1}{2}, 0)$$

$$\text{Coordinates of N } = (3, 0)$$

$$\therefore TN = 1\frac{1}{2} \text{ units} \quad (5)$$

$$a) y = \frac{x}{x-1}; x+y+5=0$$

$$y = -x-5$$

$$m = -1$$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$= \frac{x-1-x}{(x-1)^2}$$

$$= \frac{-1}{(x-1)^2}$$

$$m_{\text{tangent}} = m_{x+y+5=0}$$

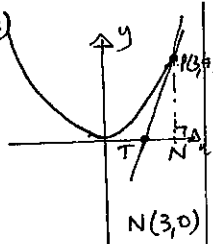
$$\frac{-1}{(x-1)^2} = -1$$

$$(x-1)^2 = 1$$

$$x-1 = \pm 1$$

$$x = 0, 2$$

$$\text{when } x=0, y=0 \therefore \text{At } (0, 0) \text{ and } (2, 2)$$



$$b) y = x^3 - x^2$$

$$\frac{dy}{dx} = 3x^2 - 2x; m = \tan 45^\circ$$

$$m = 1$$

$$\therefore 3x^2 - 2x = 1$$

$$3x^2 - 2x - 1 = 0$$

$$(3x+1)(x-1) = 0$$

$$x = -\frac{1}{3}, 1$$

$$\text{When } x = -\frac{1}{3}, y = -\frac{4}{27}$$

$$x = 1, y = 0$$

$$\therefore \text{at } (1, 0) \text{ and } (-\frac{1}{3}, -\frac{4}{27}) \quad (4)$$

Question 5 (10 marks)

$$a) y = x^3 - 3x + 1$$

$$E_{\text{tangent}}: y = mx + b$$

$$\frac{dy}{dx} = 3x^2 - 3$$

$$\text{at } x = -2$$

$$m = 3(-2)^2 - 3$$

$$= 9$$

$$E_{\text{tangent}}: y + 1 = 9(x + 2)$$

$$y + 1 = 9x + 18$$

$$y = 9x + 17$$

$$\therefore m = 9, b = 17 \quad (4)$$

$$b) y = 2x^3 + 3x - 1 \text{ at } x = 1$$

$$y(1) = 2(1)^3 + 3(1) - 1 = 4$$

$$\frac{dy}{dx} = 6x^2 + 3$$

$$= 6(1)^2 + 3$$

$$\text{at } (1, 4); m_{\text{tangent}} = 9$$

$$E_{\text{tangent}}: y - 4 = 9(x - 1)$$

$$y - 4 = 9x - 9$$

$$9x - y - 5 = 0$$

$$m_{\text{normal}} = -\frac{1}{9} \text{ pt } (1, 4)$$

$$E_{\text{normal}}: y - 4 = -\frac{1}{9}(x - 1)$$

$$9(y - 4) = -x + 1$$

$$9y - 36 = -x + 1$$

$$x + 9y - 37 = 0 \quad (5)$$

Question 6 (13 marks)

$$a) \text{ roots: } 1 + \sqrt{3}, 1 - \sqrt{3}$$

$$\alpha + \beta = 2$$

$$\alpha\beta = -2$$

$$\text{Equation: } x^2 - 2x - 2 = 0 \quad (2)$$

$$b) (2m+1)x^2 + 2mx - 1 = 0$$

$$\text{Equal roots } \Rightarrow b^2 - 4ac = 0$$

$$(2m)^2 - 4(2m+1)(-1) = 0$$

$$4m^2 + 8m + 4 = 0$$

$$m^2 + 2m + 1 = 0$$

$$(m+1)^2 = 0$$

$$m = -1 \quad (3)$$

$$\therefore \text{equal roots when } m = -1$$

$$c) x^2 - 5x + 4 = 0$$

$$i) \alpha + \beta = -\frac{b}{a} = 5 \quad (1)$$

$$ii) \alpha\beta = \frac{c}{a} = 4 \quad (1)$$

$$iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 25 - 8$$

$$= 17 \quad (2)$$

$$iv) \frac{1}{\alpha} + \frac{1}{\beta}$$

$$= \frac{\alpha + \beta}{\alpha\beta} = \frac{5}{4} \quad (2)$$

$$v) (\alpha + 2)(\beta + 2)$$

$$= \alpha\beta + 2\alpha + 2\beta + 4$$

$$= \alpha\beta + 2(\alpha + \beta) + 4$$

$$= 4 + 2(5) + 4 \quad (2)$$

$$= 18$$

Question 7 (15 marks)

$$a) 3x^2 + 8x - k$$

$$\text{positive definite } \Rightarrow a > 0, \Delta < 0$$

$$a = 3 > 0$$

$$b^2 - 4ac < 0$$

$$64 + 4(3)k < 0$$

$$12k < -64$$

$$k < -5\frac{1}{3} \quad (3)$$

$$b) x^2 - 5x + (k-1) = 0$$

$$i) \text{two real roots } \Rightarrow \Delta > 0$$

$$b^2 - 4ac > 0$$

$$25 - 4(k-1) > 0$$

$$25 - 4k + 4 > 0$$

$$-4k > -29$$

$$k < 7\frac{1}{4} \quad (3)$$

$$ii) \text{one real root}$$

$$\Delta = 0$$

$$\text{i.e. } k = 7\frac{1}{4} \quad (1)$$

$$iii) \text{no real roots}$$

$$\Delta < 0$$

$$\text{i.e. } k > 7\frac{1}{4} \quad (1)$$

$$\begin{aligned}
 c) \quad 4x^2 - 12x + 9 &\equiv A(x-1)^2 + B(x-1) + C \\
 &= A(x^2 - 2x + 1) + Bx - B + C \\
 &= Ax^2 - 2Ax + A + Bx - B + C \\
 &= Ax^2 - (2A - B)x + (A - B + C)
 \end{aligned}$$

$$\therefore A = 4$$

$$2A - B = 12$$

$$8 - B = 12$$

$$-B = 4$$

$$B = -4$$

$$A - B + C = 9$$

$$4 + 4 + C = 9$$

$$C = 1$$

(4)

$$\therefore A = 4, B = -4, C = 1$$

$$d) \quad 3^{2x} - 10 \cdot 3^x + 9 = 0$$

$$\text{let } u = 3^x$$

then

$$u^2 - 10u + 9 = 0$$

$$(u-9)(u-1) = 0$$

$$u = 1, 9$$

$$\text{i.e. } 3^x = 1 \quad \text{or} \quad 3^x = 9$$

$$3^x = 3^0 \quad 3^x = 3^2$$

$$\therefore x = 0, 2$$

(3)